

PH 101
Tutorial 6
Date: 16/09/2023

1. The system shown in Fig. 1 consists of two particles P_1 and P_2 connected by a light inextensible string of length a . The particle P_1 is constrained to move along a fixed smooth horizontal rail, and the whole system moves under uniform gravity in the vertical plane through the rail.

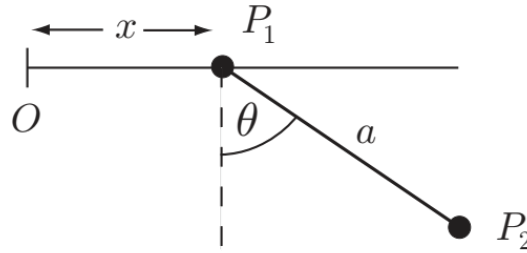


Fig. 1

For the case in which the particles are of equal mass m , show that the Lagrangian is

$$L = \frac{1}{2}m(2\dot{x}^2 + 2a\dot{x}\dot{\theta} + a^2\dot{\theta}^2) + mga \cos \theta$$

where x and θ are the coordinates shown in Figure 1. Find the Hamiltonian and verify that it satisfies the Hamilton's equations.

2. Consider the functional

$$S[\vec{q}(t), \vec{p}(t)] = \int_{t_0}^{t_1} (H(\vec{q}(t), \vec{p}(t), t) - \vec{q} \cdot \vec{p}) dt$$

of the $2n$ independent functions $\vec{q}(t) [= (q_1(t), q_2(t)) \dots q_n(t)]$ and $\vec{p}(t) [= (p_1(t), p_2(t)) \dots p_n(t)]$. Show that the extremals of S satisfy Hamilton's equations with Hamiltonian H .

3. A massless spring of length b and spring constant k connects two particles of masses m_1 and m_2 . The system rests on a smooth table and may oscillate and rotate.
 - (a) Determine Lagrange's equations of motion.
 - (b) What are the generalized momenta associated with any cyclic or ignorable coordinates?
 - (c) Determine Hamilton's equations of motion.

4. Two massless sticks of length $2r$, each with a mass m fixed at its middle, are hinged at an end. One stands on top of the other, as shown in Fig. 2.

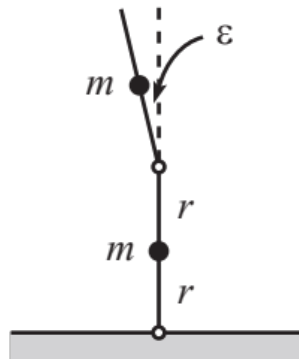


Fig. 2

The bottom end of the lower stick is hinged at the ground. They are held such that the lower stick is vertical, and the upper one is tilted at a small angle ϵ with respect to the vertical. They are then released. At this instant, what are the angular accelerations of the two sticks? **Work in the approximation where ϵ is very small.** Find the Hamiltonian under that approximation. Is H conserved? [Use small angle approximation: $\sin \theta \approx \theta$; $\cos \theta \approx 1 - \frac{\theta^2}{2}$.]